PROJECT REPORT

BREAST CANCER PREDICTION SYSTEM

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1. Abstract

This project demonstrates a supervised machine-learning workflow for classifying records in the Breast Cancer Wisconsin Diagnostic dataset as benign or malignant. It compares Logistic Regression and Random Forest classifiers and provides an interactive Streamlit interface. The project is intended for learning and demonstration only; it is not a clinical diagnostic system.

2. Objectives

- Understand a real-world binary classification dataset.
- Perform a stratified training and testing split.
- Train and compare Logistic Regression and Random Forest.
- Evaluate models using accuracy, precision, recall and F1-score.
- Build a simple web interface to demonstrate model predictions.

3. Dataset

The project uses the Breast Cancer Wisconsin Diagnostic dataset bundled with Scikit-learn. It contains 569 observations, 30 numerical input features and two classes. The target labels are benign and malignant. The training script exports a CSV copy to the dataset folder.

4. Tools and Technologies

Python, Pandas, NumPy, Scikit-learn, Streamlit, Matplotlib, Seaborn and Joblib are used in the project. The main classification models are Logistic Regression and Random Forest.

5. Methodology

1. Load the dataset and inspect the input features and target.
2. Split the data into 80% training and 20% testing sets using stratification.
3. Train Logistic Regression with feature standardization.
4. Train a Random Forest classifier with 300 trees.
5. Compare both models using held-out test-set metrics.
6. Save the model with the higher F1-score and display its metrics in Streamlit.

6. How to Run

Install dependencies with `pip install -r requirements.txt`. Train models using `python train_model.py`. Start the interface using `streamlit run app.py`. The app runs locally and displays a local address in the terminal.

7. Results

The training script prints a classification report and confusion matrix for each model and writes the comparison metrics to models/metrics.json. Run the script before final submission and record the actual values from that file here. Do not invent or estimate results.

8. Limitations and Responsible Use

This is an educational exercise using a public, pre-packaged dataset. Dataset performance does not establish clinical validity, fairness, or suitability for real patients. The application must not be used to diagnose, screen, or guide treatment. Medical decisions require qualified clinicians and validated clinical tools.

9. Conclusion

The project demonstrates the end-to-end steps of a basic supervised-learning classification task, from dataset loading and model comparison to deployment of a small interactive Streamlit application. It provides a foundation for further study of machine learning, model evaluation and responsible AI.

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